



CM NovaTouch TKL

Cooler Master unveiled the NovaTouch TKL mechanical keyboard with sound dampeners at PAX East 2014. <http://dgit.in/NovaTouch>



Logitech G502 Proteus Core

The G502 Proteus Core by Logitech is a new gaming mouse from Logitech with a 12,000 DPI sensor. <http://dgit.in/ProteusCore>

Bazaar

AMD Kaveri A8-7600

Three generations down AMD is making some progress albeit slowly

Bulldozer was a massive disappointment with all the hype around it, then came Piledriver with a few of the kinks sorted out and now it's up to Steamroller to take things forward. Bulldozer, Piledriver and Steamroller are all terms used to label each of AMD's processor microarchitecture. Another thing we'd like to remind you is about the new core-thread structure that they introduced with bulldozer wherein they re-defined the word core for their processors. Each module now shares the floating point pipeline and can handle two threads, so the A8-7600 has two modules and four threads.

The introduction of this sharing concept was quite surprising back then and everyone went after AMD for trying to cover up the processor's drawbacks by coining new terminology but it seems they had other long term plans which have come to fruition in Steamroller. Enter HSA (Heterogenous Systems Architecture) which allows both the CPU and GPU part of the APU to access the same memory. Whenever data has to be shared between the CPU and the GPU, it first has to move from, let's say, the CPU's memory to the GPU's memory where certain operations are performed after which the data is brought back into the CPU's memory. This procedure takes up precious time which can be avoided if the CPU and GPU of the APU were to be able to access the same memory space. HSA makes this possible and they are calling it hUMA (heterogenous Unified Memory Access). However, don't expect to instantly get a massive boost in performance with this new addition. Applications have to be programmed to make use of HSA. Then arises the question as to whether programmers will take to the new approach?

A new chipset and a new socket have been introduced with Steamroller, the A88X and the FM2+ socket, respectively. The FM2+ socket makes use of two extra pins, this means that the new Kaveri processors will need new motherboards and older processors will still be compatible with the new motherboards as long as the BIOS supports the use of the same.

The TDP of the processors are configurable, so if your motherboard allows modifying the TDP of your Kaveri processor then you can reduce power usage by sacrificing the processor's base clock a little. The A8-7600 allows for the TDP to be dropped from 65W to 45W dropping the base clock

from 3300 MHz to 3100 MHz. The feature wasn't polished in the initial Kaveri launches but with the A8-7600 it works much better. If the TDP were to be dropped by 5W then one can expect a performance reduction of about 6% in processor intensive benchmarks, otherwise in casual games and software that don't need much processing power, the difference is hardly felt.

When it boils down to performance, Kaveri does better than Intel processors when it comes to graphics though not to the extent that you'd be gaming effortlessly on 1080p at maximum configuration. The A8-7600 comes with an R7 GPU on the die, which has 384 cores clocked at 720 MHz. With this the APU is capable of running current generation games like Bioshock Infinite and Tomb Raider at low settings comfortably with some stuttering in certain scenarios. But reducing the resolution does improve performance to ensure a playable experience. So compared to AMD processors, this one sits somewhere between the A10-6800K and the A10-5800K while in comparison to Intel processors it performs better than HD 4600.

There are very few applications that can make use of the Heterogenous Systems Architecture that the new Kaveri processors have. LibreOffice is one such software and then there is VLC which uses OpenCL to optimise certain processes. Overall, this processor isn't something that you'd go for if CPU processing is what you need, most i5 processors are competitively priced but their graphical prowess isn't that good. Wait till we test the higher SKUs of Kaveri for more insight into the matter.

Mithun Mobandas



Performance..... 66
Value 70

Specifications

Core Name: Kaveri;
Architecture: Steamroller;
Socket: FM2+; **L2 Cache:** 2x2 MB; **Graphics:** R7; **GPU Cores:** 384; **GPU Clock:** 720 MHz; **Max DDR3 clock speed:** 2133; **Warranty:** 3 years

Contact

Company: AMD
Phone: 124 469 6000
Email: Chandrahas.Panigrahi@amd.com